A Zero-Parameter Framework for Cosmological Dynamics and Galaxy Cluster Mass Discrepancies via Structural Self-Energy Expansion (SSEE-V3.6)

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Abstract

We present the Structural Self-Energy Expansion (SSEE-V3.6), a minimalist theoretical framework that unifies cosmological expansion and cluster dynamics using dimensionless geometric constants (π and the golden ratio Φ) without introducing free parameters or exotic dark-matter particles. The framework defines a closed algebraic system yielding an asymptotic structural viscosity factor $\mathrm{KAL}_0 \approx 5.5214$, natively reproducing the observed cosmic dark-to-baryonic matter density ratio. At galactic scales, SSEE derives an acceleration threshold $a_0 = c\,H_0/\mathrm{KAL}_0 \approx 1.186 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. On cosmological scales, the model predicts a dynamical dark-energy equation of state with $w_0 \approx -0.840$ and $w_a \approx -0.670$, providing a parsimonious alignment with dynamical trends in the DESI DR2 observations [Abdul-Karim et al., 2025]. Incorporating an empirical light-neutrino mass sum $\sum m_\nu = 0.085\,\mathrm{eV}$, the framework structurally resolves the megaparsec cluster mass discrepancy when combined with a variable Integrated Galaxy-wide Initial Mass Function (IGIMF) [Zhang et al., 2026]. A full Bayesian MCMC validation against DESI DR2, Planck 2018, and galaxy-cluster mass data is presented in the companion Paper 2 [Almeida Vallejo, 2026].

1 Introduction and Theoretical Foundations

The SSEE framework operates under the principle of algebraic closure and macroscopic structural self-energy. Rather than adding particle degrees of freedom, the system's identity is defined by a closed set of geometric invariants derived exclusively from π and Φ. These fundamental scales dictate the macroscopic behaviour of spacetime:

- $\boldsymbol{\Omega}$ ($\pi + \Phi \approx 4.7596$): Stability Metric.
- $\boldsymbol{\beta}$ ($(\pi + \Phi)/2 \approx 2.3798$): Base Coupling Scalar.

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- $\mathbf{KAL_0}$ ($\beta + \pi \approx 5.5214$): Asymptotic Structural Viscosity.
- $\boldsymbol{P}$ ($\Omega + \Phi \approx 6.3776$): Dynamical Evolution Scalar.
- $\boldsymbol{K_v}$ ($\Phi + \pi + \Omega \approx 9.5192$): Structural Constraint Invariant.
- $\boldsymbol{T_r}$ ($3(\Phi + \beta) \approx 11.9935$): 3D Saturation Horizon.
- $\boldsymbol{M_v}$ ($\Phi + \pi + K_v \approx 14.2788$): Maximal Dimensional Invariant.

1.1 Axiomatic Principles

SSEE is explicitly constructed as a top-down theoretical ansatz. Its internal symmetries are not phenomenological fits derived bottom-up, but fundamental axiomatic identities that the macroscopic universe must obey. Three physical postulates govern the interactions of these geometric invariants:

1. **The Temporal Triad (Cosmological Epochs):** Spacetime evolves through three distinct thermodynamic states — the Origin/Law phase (initial singularity), the Ignition/Processing phase (structure formation), and the Transcendental Saturation phase (dark energy asymptotic domination). The Hubble history $H(a)$ is bounded by:
$$\lim_{a\to 0} H^2(a) \propto \rho_{\rm bar}, \qquad \lim_{a\to\infty} H^2(a) = \frac{8\pi G}{3}\,\rho_{\rm crit,0}\left(\frac{T_r}{M_v}\right). \tag{1}$$
2. **Non-Self-Addition (Evolution by Difference):** Structural evolution proceeds through non-linear geometric scaling rather than linear superposition. All macroscopic observables $\mathcal{O}$ must be strictly functional dependencies of the invariant set and H_0; no additional free parameters are permitted:
$$\mathcal{O} = f(\Omega,\, \beta,\, \mathrm{KAL}_0,\, P,\, K_v,\, T_r,\, M_v;\, H_0). \tag{2}$$
3. **Asymmetric Equivalence:** Energy states can possess identical geometric values but divergent dynamic functions. The macroscopic geometric response must inversely mirror the physical source:
$$\mu_{\rm SSEE}(x) \cdot \mathrm{KAL}(x) \equiv 1. \tag{3}$$

1.2 The ssee Effective Action

Instead of an arbitrary cosmological constant Λ, the vacuum is governed by a static geometric pressure field tied to the critical density $\rho_{\rm crit,0} = 3H_0^2/(8\pi G)$:
$$S_{\rm SSEE} = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} - \rho_{\rm crit,0}\left(\frac{T_r}{M_v}\right) + \mathcal{L}_{\rm matter}\right]. \tag{4}$$
This action guarantees that the geometric constraints are not phenomenological fits, but inevitable consequences of the universe's macroscopic structural self-energy at saturation.

Falsifiability. The framework is strictly falsifiable. It would be refuted by: (1) direct detection of a collisionless dark-matter particle accounting for the missing mass; (2) definitive cosmological constraints proving $w_a = 0$; or (3) high-significance confirmation of a normal neutrino hierarchy with $\sum m_\nu < 0.06\,\mathrm{eV}$.

2 Effective Field Theory Embedding

2.1 The SSEE Action

The SSEE dark-energy sector is described by a k-essence scalar field ϕ minimally coupled to gravity and extended by a geometric bulk-viscosity term in the Eckart approximation [**??**]:

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} + P(X) - \zeta \big(\nabla_\mu u^\mu\big)^2 + \mathcal{L}_m \right], \tag{5}$$

where $X \equiv -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \, \partial_\nu \phi$ is the canonical kinetic term. The fluid four-velocity of the dark-energy component is identified with the normalised gradient of the field,

$$u_\mu = -\frac{\partial_\mu \phi}{\sqrt{2X}}, \tag{6}$$

so that $u_\mu u^\mu = -1$ is satisfied automatically, and the viscous and kinetic sectors remain self-consistently defined by the single degree of freedom ϕ.

2.2 Algebraic Determination of $P(X)$

For a power-law k-essence Lagrangian $P(X) = \mathcal{A} X^n$ the equation of state is *exactly* constant:

$$w_\phi = \frac{P}{2XP_X - P} = \frac{\mathcal{A} X^n}{(2n-1)\mathcal{A} X^n} = \frac{1}{2n-1}. \tag{7}$$

Imposing the SSEE algebraic constraint $w_0 = -T_r/M_v$ uniquely fixes the kinetic exponent without any fitting:

$$\boxed{n = \frac{T_r - M_v}{2\,T_r} = \frac{1 + w_0}{2w_0} \approx -0.09527,} \tag{8}$$

where $T_r = 3(\varphi + \beta)$ and $M_v = \varphi + \pi + K_v$ are the SSEE structural constants defined in Section **??**. One verifies immediately that $1/(2n-1) = -T_r/M_v = w_0$ identically.

The overall normalisation is fixed by requiring that the present-day dark-energy density equals the SSEE prediction:

$$\rho_{\mathrm{DE},0} = \frac{3H_0^2}{8\pi G} \frac{T_r}{M_v}, \tag{9}$$

with H_0 constrained observationally to $66.66^{+0.47}_{-0.46}$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ by the MCMC analysis of Paper 2. Given the energy scale $\rho_{\mathrm{DE},0}$, the amplitude reads

$$\mathcal{A} = \frac{\rho_{\mathrm{DE},0}^{1-n}}{2n-1}, \tag{10}$$

which completes the specification of $P(X)$ with no free parameters beyond H_0.

Remark on the CPL evolution. The pure k-essence sector (**??**) delivers a *constant* $w = w_0$. The time variation $w_a = -P_{sc}/K_v$ arises from the viscous source term (Section **??**). Demonstrating that the integrated source reproduces the exact CPL form is deferred to future work.

2.3 Geometric Bulk Viscosity

The retention constant $\mathrm{KAL}_0 \equiv \beta+\pi \approx 5.521$ governs the dissipative sector. Following the Eckart first-order formalism, the bulk viscosity coefficient is coupled to the instantaneous expansion rate:

$$\zeta = \frac{\mathrm{KAL}_0}{8\pi G}\, H. \tag{11}$$

Every factor in (**??**) is either a fundamental constant (G) or an algebraic SSEE invariant (KAL_0, H via the background equations), so no new parameter is introduced.

The viscous pressure is

$$\Pi = -\zeta\, \nabla_\mu u^\mu = -3\zeta H = -\frac{3\,\mathrm{KAL}_0\, H^2}{8\pi G}. \tag{12}$$

Note that $\Pi < 0$ (negative viscous pressure), consistent with the dissipative retention of energy at high H.

Physical motivation for $\zeta \propto H$. Bulk viscosity proportional to the Hubble rate arises naturally in kinetic theory when the dominant relaxation time is set by the expansion itself. This form has been studied in the context of bulk-viscous dark energy by **?** and, in a relativistic treatment, by **?**. Within SSEE the coupling constant is not fitted but predicted geometrically as $\mathrm{KAL}_0 = (\pi + \varphi)/2 + \pi$.

2.4 Modified Friedmann Equations

Applying the principle of least action ($\delta S = 0$) to the FLRW metric $ds^2 = -dt^2 + a^2(t)\, d\mathbf{x}^2$ yields three governing equations for the SSEE background.

I. Energy equation:

$$H^2 = \frac{8\pi G}{3}\big(\rho_m + \rho_r + \rho_{\mathrm{SSEE}}\big). \tag{13}$$

II. Acceleration equation:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left[\rho_m + 2\rho_r + \rho_{\mathrm{SSEE}}(1+3w_0) - \frac{9\,\mathrm{KAL}_0\, H^2}{8\pi G}\right]. \tag{14}$$

III. Conservation equation with dissipative source:

$$\dot{\rho}_{\mathrm{SSEE}} + 3H\,\rho_{\mathrm{SSEE}}(1+w_0) = \frac{9\,\mathrm{KAL}_0\, H^3}{8\pi G}. \tag{15}$$

Derivation of the coefficient 9. From Eq. (**??**), the viscous pressure is $\Pi = -3\mathrm{KAL}_0 H^2/8\pi G$. In the Raychaudhuri equation the effective pressure of the SSEE sector enters as $p_{\mathrm{eff}} = w_0\rho_{\mathrm{SSEE}} + \Pi$, so the acceleration term becomes

$$\begin{aligned} -\frac{4\pi G}{3}\big(\rho_{\mathrm{SSEE}} + 3p_{\mathrm{eff}}\big) &= -\frac{4\pi G}{3}[\rho_{\mathrm{SSEE}}(1+3w_0) + 3\Pi] \\ &= -\frac{4\pi G}{3}\left[\rho_{\mathrm{SSEE}}(1+3w_0) - \frac{9\,\mathrm{KAL}_0\, H^2}{8\pi G}\right], \end{aligned} \tag{16}$$

which is exactly Eq. (**??**). The factor 9 arises from $3 \times |\Pi|/(H^2) = 3 \times 3\mathrm{KAL}_0/8\pi G$; a factor of 3 rather than 9 would correspond to including Π only once instead of as the trace of the viscous stress.

Similarly, the conservation law $\nabla_\mu T^{\mu\nu} = 0$ for the effective fluid (w_0, Π) gives $\dot{\rho} + 3H(\rho + w_0\rho + \Pi) = 0$, hence $\dot{\rho} + 3H\rho(1 + w_0) = 9\mathrm{KAL}_0 H^3/8\pi G$, confirming that Eq. (**??**) and Eq. (**??**) are mutually consistent.

2.5 Physical Regimes

Late universe ($z \lesssim 2$, DESI DR2). At low redshift $H \approx H_0$, so the viscous source $\propto \mathrm{KAL}_0 H^3/8\pi G$ is small compared to ρ_{SSEE}. The acceleration equation is dominated by $\rho_{\mathrm{SSEE}}(1+3w_0) < 0$ (since $w_0 \approx -0.840 < -1/3$), driving $\ddot{a} > 0$ without dark matter. This regime is the one tested against DESI DR2 in Paper 2, where the algebraic prediction (w_0, w_a) matches the data at 0.05σ.

Early universe (CMB epoch, $z \sim 10^3$). At recombination $H \gg H_0$, the geometric retention term $9\,\mathrm{KAL}_0 H^3/8\pi G$ acts as a dissipative source in Eq. (**??**), injecting energy into the dark-energy fluid and modifying its effective equation of state. This structural viscosity provides the physical mechanism by which an effective matter density $\Omega_{m,\mathrm{eff}} = 1 - T_r/M_v \approx 0.160$ yields the correct CMB acoustic scale through the MIRA factor $\mathcal{M} = 2 - 1/\varphi^2 \approx 1.9989$ (Paper 3, §2.3).

2.6 Summary and Open Problems

Table **??** summarises the EFT status. The action (**??**) with kinetic exponent (**??**) and viscosity (**??**) constitutes a *internally consistent*, zero-free-parameter EFT description of the SSEE background. Every constant (n, KAL_0, Ω_{DE}) is derived from φ and π; only the overall energy scale H_0 enters as an observational input.

Three aspects require further development:

1. **Derivation of w_a from EFT.** The viscous source in Eq. (**??**) generates a running effective EOS $w_{\mathrm{eff}}(z)$. Whether this running reproduces the exact value $w_a = -P_{sc}/K_v$ algebraically requires integrating Eq. (**??**) numerically.

2. **Cosmological perturbation theory.** The scalar power spectrum and CMB transfer functions in the presence of bulk viscosity require the full Israel–Stewart formalism [**?**].

3. **Radiative stability.** The EFT cutoff scale Λ_{UV} and the quantum stability of the $n < 0$ power-law sector are not yet established.

3 Dark Energy Evolution

SSEE models dynamical dark energy via the Chevallier–Polarski–Linder (CPL) form $w(a) = w_0 + w_a(1-a)$. Motivated by Eq. (4), the expansion is subject to an effective geometric pressure:

$$H^2 = \frac{8\pi G}{3}(\rho_{\mathrm{bar}} + \rho_{\mathrm{DE}}), \qquad \rho_{\mathrm{DE}} = \rho_{\mathrm{crit},0}\left(\frac{T_r}{M_v}\right). \tag{17}$$

Table 1: EFT status summary for SSEE-V3.6.

Element	Status	Derivation
Action form S	✓Complete	Standard GR + k-essence + Eckart
$u_\mu(\phi)$ coupling	✓Complete	$u_\mu = -\partial_\mu\phi/\sqrt{2X}$
Kinetic exponent n	✓Algebraic	$n = (T_r - M_v)/2T_r$ from φ, π
Equation of state w_0	✓Algebraic	$w_0 = 1/(2n-1) = -T_r/M_v$
Viscosity coupling ζ	✓Algebraic	$\zeta = \mathrm{KAL}_0 H/8\pi G$
Friedmann eqs. I–III	✓Consistent	Factor 9 verified in §**??**
w_a from EFT	○ Open	Requires numerical integration
Perturbation spectrum	○ Open	Requires Israel–Stewart
Radiative stability	○ Open	Future work

The baseline state w_0 is set by the static geometric ratio T_r/M_v, and the evolution w_a encodes the dynamical departure from saturation driven by P/K_v:

$$w_0 = -\frac{T_r}{M_v} \approx -0.840, \qquad w_a = -\frac{P}{K_v} \approx -0.670. \tag{18}$$

Asymptotic saturation. Although $w_0 + w_a = -1.510$ would ordinarily imply a phantom Big Rip, SSEE adopts CPL only as a local observational parameterisation. In the far future ($a \to \infty$) the energy density saturates at the finite T_r/M_v boundary of Eq. (4), ensuring universal stability.

DESI DR2 [Abdul-Karim et al., 2025] analyses indicate an evolving preference for dynamical dark energy. The SSEE algebraic point $(w_0, w_a) = (-0.840, -0.670)$ falls within the 68% confidence contours of the combined BAO+CMB+DESY5 datasets (Mahalanobis $\chi^2_{2D} = 0.080$, i.e. 0.05σ; quantified in Paper 2 [Almeida Vallejo, 2026]).

4 Galaxy Cluster Dynamics

4.1 The Cluster Mass Discrepancy and IGIMF

In rich galaxy clusters, inferred dynamical mass typically exceeds the visible baryonic mass by a factor of ∼5–6. Recent literature incorporating a variable IGIMF [Zhang et al., 2026] demonstrates that baryonic masses have been systematically underestimated: heavy stellar remnants raise the effective baryonic baseline to at least $88^{+5}_{-4}\%$ of the MOND dynamical mass. In SSEE, this updated IGIMF baseline multiplied by $\mathrm{KAL}_0 \approx 5.5214$ natively closes the remaining discrepancy without cold dark matter.

4.2 Structural Viscosity and the Neutrino Bridge

At the cluster scale, the total effective dynamical mass $M_{\rm dyn}$ within radius R is:

$$M_{\rm dyn}(R) = M_{\rm bar}^{\rm IGIMF}(R) \cdot \mathrm{KAL}_0 \cdot [1 + f_\nu]\,, \tag{19}$$

where f_ν quantifies the fractional mass contribution of trapped relic neutrinos. Using $\sum m_\nu = 0.085\,\mathrm{eV}$ — robustly permitted within SSEE's dynamic dark-energy regime — the cosmological neutrino density is $\rho_\nu \approx 28.6\,\mathrm{eV\,cm^{-3}}$. For a cluster with escape velocity $v_{\rm esc} \approx 2000\,\mathrm{km\,s^{-1}}$, integration of the Fermi–Dirac distribution [Tremaine & Gunn, 1979]

yields a trapped fraction $f_\nu \approx 0.018$–0.022. This structurally consistent $\sim$2% correction provides the final macroscopic bridge.

4.3 Quantitative Cluster Analysis

Applying Eq. (7) to four well-documented clusters demonstrates predictive accuracy against observed dynamical masses (Table 1).

Table 2: Cluster mass predictions. All masses within R_{500} in units of $10^{14}\,M_\odot$. $M_{\rm SSEE} = M_{\rm bar}^{\rm IGIMF} \times 5.5214 \times 1.02$. IGIMF baselines from Zhang et al. [2026]; observed masses from Simionescu et al. 2011, Walker et al. 2012, Pointecouteau et al. 2004, Clowe et al. 2006.

Cluster	$M_{\rm bar}^{\rm IGIMF}$ $[10^{14}\,M_\odot]$	$M_{\rm SSEE}$ $[10^{14}\,M_\odot]$	$M_{\rm dyn}^{\rm obs}$ $[10^{14}\,M_\odot]$
Coma	1.8 ± 0.2	10.1 ± 1.1	9.8 ± 1.0
A2029	2.2 ± 0.2	12.4 ± 1.1	12.0 ± 1.2
A478	1.5 ± 0.2	8.4 ± 1.1	8.0 ± 1.0
Bullet (main)	1.2 ± 0.2	6.7 ± 1.1	6.5 ± 1.0

All four predictions agree with observed masses at the 1σ level ($\chi^2_r = 0.122$; full analysis in Paper 2 [Almeida Vallejo, 2026]).

4.4 Gravitational Lensing and the Newtonian Limit

The Bullet Cluster (1E 0657-56) is frequently cited as proof of particulate dark matter due to the spatial offset between collisional X-ray gas and lensing-mass peaks [Clowe et al., 2006]. SSEE reproduces this topological offset through macroscopic structural viscosity without collisionless particles.

The structural viscosity acts as an environment-dependent scalar field. Defining $x = |\nabla\Phi|/a_0$, SSEE adopts an interpolation function analogous to the canonical MOND simple interpolating function $\mu(x)$:

$$\mu_{\rm SSEE}(x) = \frac{x+1}{x+{\rm KAL}_0} \implies {\rm KAL}(x) = \frac{1}{\mu_{\rm SSEE}(x)} = \frac{x+{\rm KAL}_0}{x+1}. \tag{20}$$

For $x \gg 1$ (Newtonian limit), ${\rm KAL}(x) \to 1$. For $x \ll 1$ (deep MONDian limit), ${\rm KAL}(x) \to {\rm KAL}_0$. During a high-velocity cluster collision, the stripped X-ray gas flattens its local potential gradient ($x \ll 1$), activating the full ${\rm KAL}_0$ amplification; collisionless galaxies retain compact potential wells ($x \geq 1$), coupling non-linearly.

The effective lensing convergence is:

$$\kappa(\vec{\theta}) = \frac{\Sigma_{\rm SSEE}(\vec{\theta})}{\Sigma_{\rm crit}}, \quad \Sigma_{\rm SSEE}(\vec{\theta}) = \int \rho_{\rm bar}^{\rm IGIMF}(\vec{\theta},\ell)\cdot {\rm KAL}(x)\,d\ell, \quad \Sigma_{\rm crit} = \frac{c^2}{4\pi G}\frac{D_S}{D_L D_{LS}}. \tag{21}$$

This allows lensing peaks to track galaxies, consistent with QUMOND simulations [Angus et al., 2007].

Data Availability

All observational data used in this work are publicly available. DESI DR2 BAO measurements are from Abdul-Karim et al. [2025] (`https://arxiv.org/abs/2503.14738`).

Galaxy-cluster dynamical masses are from Zhang et al. [2026] (https://arxiv.org/abs/2602.06082). Analysis scripts and data files are available at https://github.com/mikealmeida1721/SSEE.

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